

Lithological and structural controls on surface deformation in the mudstone-rich fold-thrust belt of Southwestern Taiwan: Insights from numerical modeling[☆]

Wan-Lin Hu^{*}, Eh Tan

Institute of Earth Sciences, Academia Sinica, Taipei, Taiwan

ABSTRACT

Understanding surface deformation mechanisms is essential for hazard assessment in fold-and-thrust belts, yet the role of non-elastic geological units remains poorly quantified compared to traditional fault-based elastic dislocation models. Southwestern Taiwan's fold-and-thrust belt exhibits anomalously high aseismic uplift and strain rates over length scales of ~ 10 km, despite sparse shallow seismicity. These observations have been attributed either to lithologically controlled non-elastic behavior of mudstones (e.g., mud diapirism) or to structural controls from fault geometry, but their relative roles remain unresolved. To evaluate these mechanisms, we conducted two-dimensional elasto-visco-plastic numerical experiments based on a representative geological profile across the Tainan sector of the fold-and-thrust belt. The models tested end-member rheological scenarios—frictional versus viscous behavior of mudstone units—and structural configurations including forethrusts, backthrusts, and detachment ramps. Our results indicate that structural architecture exerts the dominant control on the uplift pattern. In particular, the combination of active backthrusts and detachment ramps reproduces the observed asymmetric vertical-velocity pattern, with a broad, gentle western flank and a narrow, steep eastern flank, whereas lithological weakening amplifies but does not by itself generate this asymmetry. These findings provide a framework for interpreting geodetic observations and establish predictive patterns for future monitoring.

1. Introduction

Understanding the distribution of strain, including its spatiotemporal characteristics and governing mechanisms, is fundamental to seismic hazard assessment. For example, shallow portion of faults are typically interseismically locked, leading to stress accumulation over time; this drives the buildup of elastic strain, which is subsequently released during earthquakes. As a result, interseismic geodetic strain rates, when combined with elastic dislocation models, are often used to infer elastic strain buildup along faults and to estimate seismic potential (e.g., Savage, 1983; Okada, 1985). Observational evidence also shows a positive correlation between higher strain rates along faults and greater seismic activity (e.g., Stevens and Avouac, 2021). However, some studies suggest that observed geodetic strain rates may not directly translate to seismic potential, as they can reflect aseismic processes such as fault creep or responses from viscoelastic layers at depth, rather than representing unreleased elastic strain available for future earthquakes (e.g., Masson et al., 2004; Schmidt et al., 2005; Sue et al., 2007; Wang et al., 2021). These findings challenge conventional approaches to assessing seismic hazard and underscore the importance of constraining

the characteristics of strain distribution, slip behavior, and the underlying mechanisms driving surface deformation.

Elastic dislocation theory is widely used to model observed interseismic deformation associated with faults (e.g., Savage, 1983). It effectively captures first-order observations (e.g., Ching et al., 2011; Hsu et al., 2003). However, relatively little focus has been placed on modeling underground aseismic deformation (or creep), driven by non-elastic deformation due to ductile flow, and its impact on surface deformation over decadal timescales. Furthermore, the contribution of underground aseismic slip to surface deformation, whether associated with faults or the surrounding rock mass, remains insufficiently explored. The fold-and-thrust belt (FTB) in southwestern (SW) Taiwan provides an excellent case study to further investigate these topics.

1.1. The fold-and-thrust belt in southwestern Taiwan

The Taiwan orogenic belt, situated at the convergent boundary between the Eurasian and Philippine Sea plates, accommodates approximately 88 mm/yr of total convergence across the orogen (Yu et al., 1999). In southwestern Taiwan, a FTB has developed (Fig. 1a), taking

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^{*} Corresponding author.

E-mail addresses: huwanlin@earth.sinica.edu.tw (W.-L. Hu), tan2@earth.sinica.edu.tw (E. Tan).

up approximately 30 mm/yr of this shortening (e.g., Ching et al., 2016; Le Béon et al., 2017). Geologically, the FTB in SW Taiwan features extensive mudstone outcrops of the Gutingkeng (Gtk) Formation (e.g., Mouthereau et al., 2001), as mapped by Chinese Petroleum Corporation, CPC (1989) (Fig. 1b). Structurally, compared to the FTB in northwestern Taiwan, the FTB in SW Taiwan exhibits distinct structural

characteristics. These include both forethrusts and backthrusts (Fig. 1d), as well as regularly spaced anticlinal morphological patterns extending from offshore to onshore (e.g., Chen et al., 2014; Huang et al., 2004; Lacombe et al., 2004; Liu et al., 1997; Shyu et al., 2005; Sun and Liu, 1993; Wang et al., 2022) (Fig. 1a). These folds correlate with positive Bouguer gravity anomalies—indicative of underlying high-density

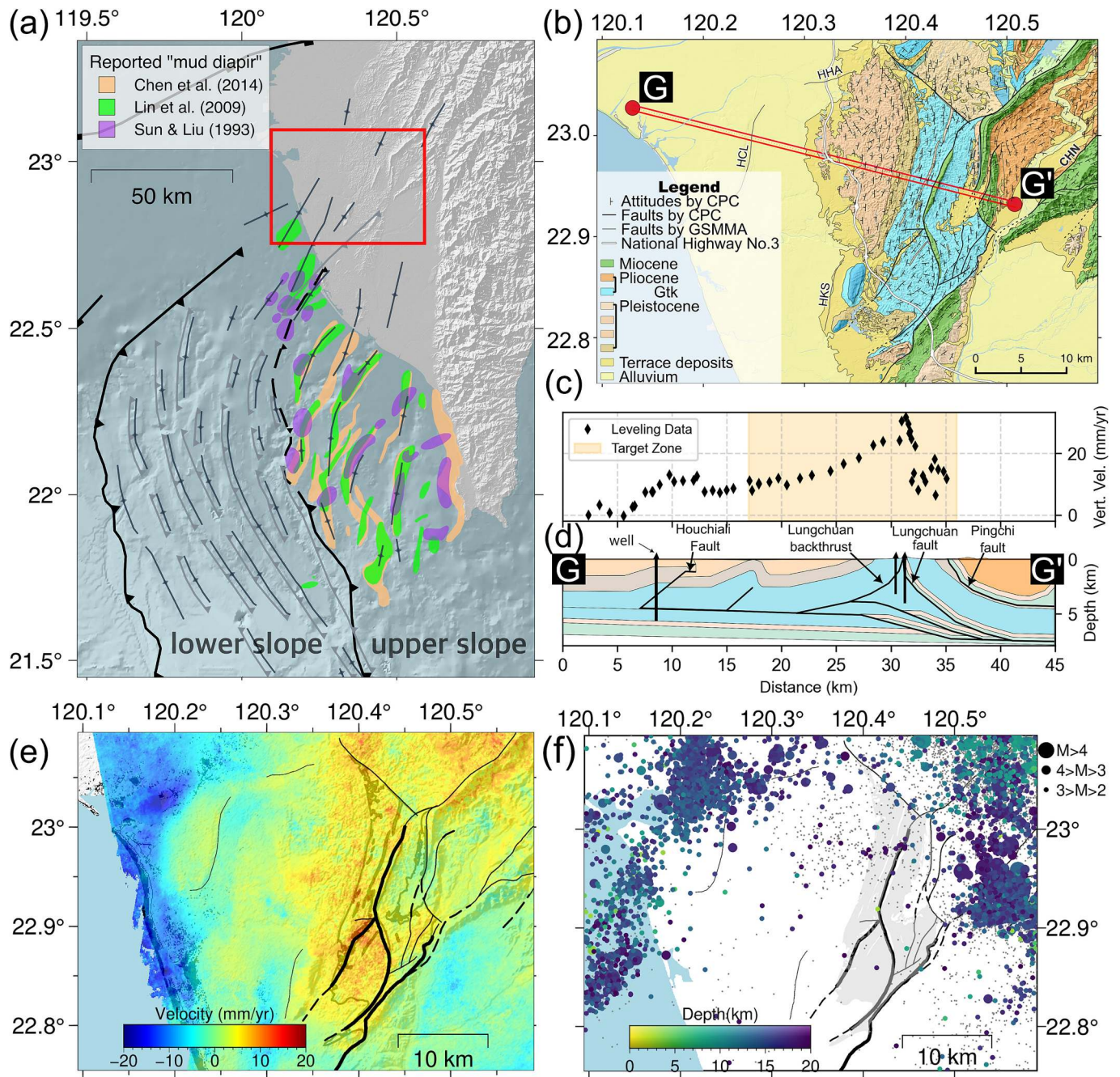


Fig. 1. Overview of the fold-and-thrust belt (FTB) in southwestern Taiwan. (a) The offshore upper slope features folding without clear thrusts and reported dome-shaped mud diapirs or mud-cored anticlines (Chen et al., 2014; Lin et al., 2009; Sun and Liu, 1993), whereas the lower slope displays a structural pattern characterized by thrust faults and folds, the latter being defined by anticlinal axes (Liu et al., 1997). These folding structures continue onshore. Faults are compiled from the Geological Survey and Mining Management Agency, Ministry of Economic Affairs, and the Chinese Petroleum Corporation (CPC) (1989). The red rectangle indicates the area corresponding to panels (b), (e), and (f). (b) Geological map after CPC (1989) and Le Béon et al. (2017). Light blue represents the mudstone-rich Gutingkeng (Gtk) Formation. In panels (c), (e), and (f), the light gray shading (or gray line) indicates the horizontal extent of the mudstone area, shown in panel (b). (c) Vertical velocity derived from leveling measurements (Le Béon et al., 2017); the highlighted data range corresponds to Fig. 11. (d) Corresponding structural model along profile G–G', modified from Le Béon et al. (2017). (e) Vertical velocity derived from InSAR data (2016–2020), corrected with GNSS data (Franklin and Huang, 2022). (f) Seismic relocations from 1990 to 2020 for earthquakes shallower than 20 km; color represents depth and size corresponds to magnitude. This catalogue is updated from Wu et al. (2008) and is available at <http://seismology.glnu.edu.tw/>. The gray filled background indicates the mudstone area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mudstone—reflecting an interplay between lithology, surface morphology, and structural development (e.g., Doo et al., 2015; Hsieh, 1972; Lo et al., 2023; Pan, 1968).

The geodetic observations reveal anomalously high vertical uplift and strain in the FTB of SW Taiwan relative to its neighboring counterparts and theoretical expectations for interseismic loading (e.g., Ching et al., 2016; Franklin and Huang, 2022) (Fig. 1c, e). Fig. 1c further shows that the uplift distribution is distinctly asymmetric, with peak vertical velocities centered over the mudstone-rich zone. The western flank exhibits a broad, gentle gradient toward the foreland, whereas the eastern flank drops off steeply toward the hinterland, highlighting localized deformation. However, this region has experienced very few shallow (<20 km) crustal earthquakes in recent decades (Central Weather Administration, CWA, Taiwan, 2012; Mouthereau et al., 2009; Wu et al., 2009; Wu et al., 2008) (Fig. 1f). This apparent mismatch between surface deformation and seismic activity, has led to the suggestion that the region may harbor significant seismic potential (Chan et al., 2020). However, geological and geodetic studies both indicate that a large component of aseismic deformation occurs during the interseismic period (Ching et al., 2011; Fruneau et al., 2001; Huang et al., 2006; Le Béon et al., 2024; Le Béon et al., 2019; Le Béon et al., 2017). This makes the SW Taiwan FTB a critical region to study how strain is accommodated by aseismic deformation and for refining assessments of long-term seismic hazard.

Recent studies suggest that interseismic and co-seismic deformation cannot be fully explained by shallow crustal faults alone or the deeper seismogenic fault (e.g., Ching et al., 2016; Huang et al., 2016; Lin et al., 2024; Mai et al., 2021; Rau et al., 2022; Tsukahara and Takada, 2018). Given the high aseismic deformation rates observed over spatial scales of less than 10 km and their correlation with mudstone exposures, mud diapirism—implying lithologically controlled non-elastic behavior—has been proposed as a mechanism to explain such anomalously high strain rates and observed deformation patterns (e.g., Ching et al., 2016; Lin et al., 2024; Mai et al., 2021).

Several observations worldwide appear to support these hypothesized mechanisms. For instance, lithified outcrops and seismic images reveal that shale or mudstone substrates often form diapiric structures distinct from those of purely brittle layers (e.g., Dinc et al., 2022; Hudec and Soto, 2021; Morley et al., 2017). Similarly, discussions have emerged in other fold-and-thrust belts, such as the Niger Delta, regarding whether long-term folding morphologies result from ductile deformation within shale substrates or, as conventionally interpreted, fault-dominated brittle deformation (Restrepo-Pace, 2020). Additionally, mudstone or shale appear to exhibit markedly different mechanical properties, with their mobility increases drastically across critical states (e.g., Soto et al., 2021; Yassir, 2003). However, adapting these mechanisms—suggested to be associated with long-term processes—to explain decadal observations (e.g., Ching et al., 2016) has yet to undergo comprehensive evaluations of the mechanical properties of faults and rock layers or the incorporation of realistic fault geometry.

1.2. Research objectives and methodology

In this study, we aim to investigate controlling factors of surface deformation in the FTB of SW Taiwan through numerical geodynamic models. We focus on a specific section where the geodetic observations (Fig. 1c) are spatially coincident with the geological profile (Fig. 1d) proposed by Le Béon et al. (2017), which was constructed based on the geological map produced by CPC (1989) (Fig. 1b). We develop a model that incorporates key structural features. Each lithological and structural unit has its own mechanical properties, i.e., friction and viscosity parameters. We vary the mechanical parameters of each unit to identify their respective impacts on deformation pattern, such as the strain rate distribution, surface deformation pattern, and surface velocity. This approach enables us to explore how variations in subsurface rheology and fault geometry influence surface observations.

Our numerical approach is based on solving the Stokes equation, which governs the behavior of slow, creeping flows in the lithosphere and ensures that stresses are in mechanical equilibrium. In this framework, surface deformation emerges as a direct consequence of internal stress distributions, which are themselves strongly influenced by the geometry and spatial distribution of mechanical structures such as faults, rheological interfaces, and material heterogeneities. However, both the geometry and rheology of these structures are subject to considerable uncertainty, and importantly, their effects on surface deformation are strongly coupled. In this study, we constrain the geometry using a geologically plausible structural profile and systematically vary rheological properties to isolate their influence on modeled deformation. By explicitly resolving these features in the model domain, we can capture their control on stress localization and deformation patterns.

This challenge of disentangling the effects of loading geometry (i.e., the spatial distribution, thickness, and lateral extent of surface mass, such as ice sheets, sediments, or water bodies) and rheology from surface deformation is well-established in Earth science applications, particularly in studies of glacial isostatic adjustment (GIA), where the surface response to loading and unloading of ice is typically modeled using the Stokes formulation (e.g., Yuan and Zhong, 2025). In GIA models, both the ice loading history and the Earth's viscosity structure are not uniquely constrained in isolation but, instead, are iteratively updated in tandem to achieve consistency between modeled and observed deformation. (e.g., Paulson et al., 2007). This coupling reflects the inherent interdependence between the imposed forcing (ice load) and the Earth's rheological response. A similar coupling of uncertainties arises in our study, where surface deformation is jointly governed by the structural geometry and the effective rheology of subsurface units.

Similarly, while surface geology in SW Taiwan provides strong evidence for the presence of key structural units, both the precise geometry at depth and the effective rheological properties of these units remain poorly constrained. Since both factors significantly influence the geodetically observed surface deformation, they cannot be uniquely resolved. To address this, we adopt a physics-based framework analogous to GIA studies, where loading geometry and rheology are iteratively refined to match observed deformation.

This study serves as a first step, aimed at evaluating the sensitivity of surface deformation to end-member assumptions about subsurface rheology. Building on insights gained from this initial phase, future models will systematically vary the structural geometry, such as the dips of the thrusts and the depth of the detachment, within bounds of the geological constraints. Moreover, the current 2-D configuration does not capture along-strike variations known to exist in this region (e.g., Aleem et al., 2025). Future work will incorporate these lateral variations and refine both geometry and rheological parameters. Through this process, we aim to progressively narrow the range of plausible subsurface models and better constrain the mechanical behavior of the FTB in southwestern Taiwan.

2. Methodology

2.1. DynEarthSol2D method

To solve the problem of stress and strain distribution, this research employs DynEarthSol2D (DES) (Choi et al., 2013), an open-source numerical modeling code specifically designed for geodynamic problems. By employing an explicit unstructured (adaptive triangular) finite element solver, DES enables the use of an unstructured mesh, allowing for varied spatial resolution in designated areas, thereby enhancing the model's ability to capture localized features. DES operates on Lagrangian meshes to solve the quasistatic momentum equilibrium via the momentum equation:

$$\rho \dot{\mathbf{u}} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g} \quad (1)$$

where ρ is the scalar material density, $\dot{u}$ is the velocity time derivative vector (acceleration), σ is the stress tensor (rank-2 tensor), g is the gravitational acceleration vector, and $\nabla \cdot$ is the divergence operator. The full stress tensor is decomposed into deviatoric and volumetric components following the constitutive relationship:

$$\sigma = \sigma' - P\mathbf{I} \quad (2)$$

where σ' is the deviatoric stress tensor (rank-2 tensor), P is pressure (scalar), and $\mathbf{I}$ is the identity tensor (rank-2 tensor).

The mechanical behavior is governed by an elasto-visco-plastic constitutive law. Before yielding, the material deforms as a visco-elastic solid. The deviatoric stress evolves according to the relationship between the shear modulus G , the viscosity (scalar) η , and the deviatoric strain rate tensor (rank-2 tensor) $\dot{\epsilon}'$:

$$\dot{\sigma}' = 2G\dot{\epsilon}' - \frac{G}{\eta}\sigma' \quad (3)$$

The plastic behavior follows a Mohr-Coulomb yield criterion:

$$\tau_c = \sigma_n \tan \phi_0 + C_0 \quad (4)$$

Eq. (4) describes the relationship between the critical shear stress (τ_c) required to cause failure in a material, the normal stress (σ_n), the coefficient of internal friction ($\tan \phi_0$), and the cohesive strength (C_0). For the viscous component, the viscosity (η) is governed by a non-Newtonian dislocation creep law as defined by [Chen and Morgan \(1990\)](#):

$$\eta = \frac{1}{4} \left(\frac{3}{4A} \right)^{\frac{1}{n}} (\epsilon_{II})^{\frac{1-n}{n}} \exp \left(\frac{E}{nR(T+273)} \right) \quad (5)$$

where ϵ_{II} is the second invariant of the deviatoric strain rate tensor; R is the universal gas constant; T is the temperature in Celsius; and n , A , E , defined as scalar values, are specific viscosity parameters assigned in the model to capture the appropriate viscosity behavior for the problem being studied.

DES has been successfully applied in modeling the long-term evolution of fault zones with finite widths and structures in fold-and-thrust belts (e.g., [Feng et al., 2015](#); [Tong and Lavier, 2018](#); [Lee et al., 2024](#)). Unlike previous studies that focus on long-term evolution, this research employs DES to investigate surface deformation and its driving mechanisms in decadal timescales. At this timescale, the initial structural geometry is assumed to remain largely unchanged. Under these conditions, we analyze surface responses influenced by specific mechanical properties where the ambient stress conditions increase with progressive boundary compression. This approach enables comparison with current observations under a range of possible in-situ stress conditions.

2.2. Model configuration

Here, we describe the model configuration. The model domain is 100 km in width and 10 km in depth, with the area of interest centered between $X = 37$ and 67 km. The resolution is 0.2 km in the central region. A shortening rate of 30 mm/yr is applied at the right-side boundary, based on present-day shortening rate ([Le Béon et al., 2017](#)). The bottom and left boundaries are impermeable in normal direction and free slip in shear direction. The top boundary is free surface with a hill-slope diffusion for the surface processes. Since surface processes are not the primary focus of this study, a constant diffusion coefficient of $5 \times 10^{-6} \text{ m}^2/\text{s}$ (e.g., [Choi et al., 2013](#)) is used across all models. We assume a surface temperature of 20 °C and an initial geothermal gradient of approximately 25 °C/km.

For modeling the present-day structural profile, we constructed our model based on the structural geometry derived from balanced cross-sections informed by surface geology and borehole drilling data ([Fig. 1d](#), [Le Béon et al., 2017](#)). The structural framework includes the Lungchuang forethrust, backthrust, a detachment with a ramp-and-flat

geometry, surrounding rock masses with sandstone substrates and multiple units of the Gutingkeng (GtK) Formation, which predominantly consists of mudstone ([Fig. 1b, c](#) and [Fig. 2](#)). The model space is divided into 10 units ([Fig. 2b](#)), each representing either a structural unit (faults), composed of a narrow zone, or a lithological unit. Each unit has its own mechanical properties. Units dominated by mudstone are designated as “M”. Because the central unit of the GtK Formation is documented to be under high pore-fluid pressure ([Yuan et al., 1987](#)), the mechanical properties of the central unit likely differ from those of the flank units. These GtK units are assigned with individual labels. The central unit is labeled M1, while the two flank units are designated as M2. The remaining host rock units are primarily composed of sandstone, labeled as ss. Since these host rock units do not deform significantly, their mechanical properties were not specifically analyzed in this study. Due to the limitations of finite element modeling in representing fault discontinuities, fault zones were modeled as structural units with ~0.5 km thickness. The three structural units—forethrust, backthrust, and detachment—are labeled T1, T2, and T3, respectively ([Fig. 2](#)). The lower friction angle (i.e., 8°, 16°) adopted in the model represents mechanical behavior under higher pore pressure conditions, where pore pressure approaches lithostatic values and the effective normal stress is correspondingly reduced. In contrast, near the surface, the pore pressure is generally close to hydrostatic, and thus near-surface materials are not expected to exhibit such low friction angle. To prevent unrealistically high uplift rates caused by low-friction materials reaching the surface, a 0.5 km-thick elastic cap layer with properties identical to the surrounding host rock is incorporated near the surface.

Since we focus on creep behaviors, all units are assigned identical elastic moduli, as these parameters play a secondary role. The bulk and shear moduli are set at 50 GPa and 30 GPa, respectively. Cohesion values are assigned to strictly isolate the mechanical behavior of specific structural units. Targeted units are assigned zero cohesion to facilitate yielding, while non-targeted units are assigned extremely high cohesion to mechanically inhibit deformation. Although natural rocks are not perfectly rigid, this high cohesion contrast is necessary to prevent strain from partitioning into the background matrix. This ensures that strain localization is driven exclusively by the geometry and motion of the target unit, even when that unit is rheologically strong. The fault zones were given a density of 2200 kg/m³, lower than the surrounding host rock layers (ss) (2500 kg/m³ for the Late Miocene Basement, based on [Hsieh \(1970\)](#)). The target mudstone formation is assigned a density of 2700 kg/m³, slightly higher than the surrounding rock layers. This assignment references [Lo et al. \(2023\)](#), who observed slightly higher gravity anomalies (1–2 mGal, corresponds to ~200 kg/m³) in mud-core anticlinal areas. Thus, using a higher density for the mudstone is reasonable for M1 and M2.

Specific mechanical properties were assigned to each unit in the models. In each model, we test specific units (i.e., M1, M2, T1, T2, and T3), hereafter referred to as “active” units—that is, units that are targeted and permitted to yield and deform—and the model is named after the tested unit(s). If the model tests multiple units, it is named using the combined unit labels (e.g., T123 indicates that T1, T2, and T3 units are tested in the model). Each tested unit behaves either viscously or plastically, but not both. To ensure exclusive behavior, viscous units have low viscosity with extremely high friction and vice versa. In experiments focusing on **plastic** behavior, the friction angle was varied systematically from 32°, 24°, 16°, 8° (corresponding to frictional coefficients of 0.62, 0.45, 0.29, 0.14). The unit label in the model's name is suffixed with ‘f’ accordingly. For the viscous behavior, the viscosity parameters (n , A , E) are varied. We consider the following viscosity profiles: constant, thermal-activated Newtonian viscosity, and thermal-activated non-Newtonian, which are denoted by the suffixes ‘cv’, ‘nv’, and ‘nnv’, respectively ([Fig. 3](#)). Each profile has 4 different variations. To isolate the effects of specific structural units and their mechanical properties on surface velocity, the mechanical properties of all non-targeted units were set to be infinitely strong, effectively rendering

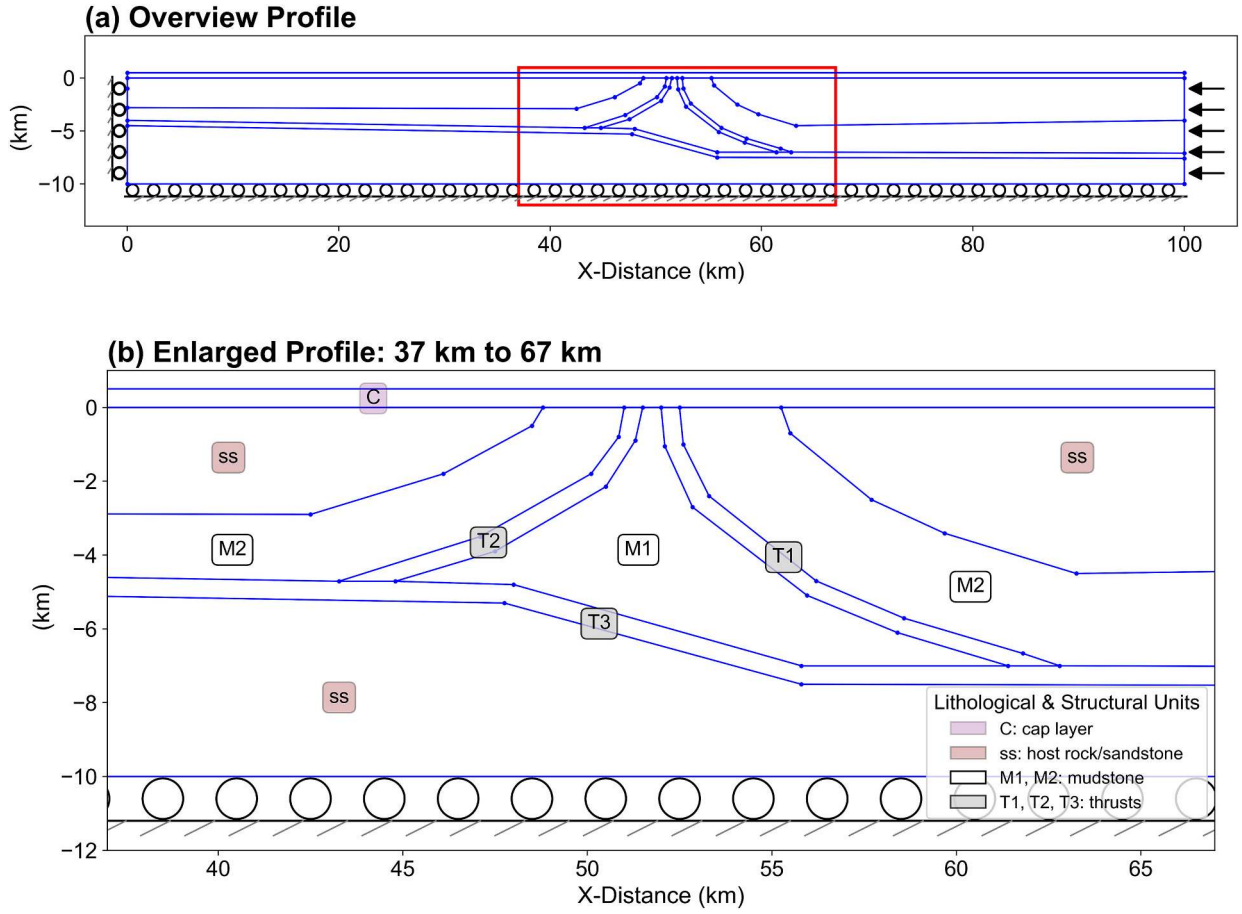


Fig. 2. Overview of the 2D model space showing (a) the full profile and (b) an enlarged section from $X = 37$ to 67 km (indicated by the red rectangle in panel (a)). Structural and lithological units are adapted from Le Béon et al. (2017). Lithological units: M1 represents the high-pressure zone between the forethrust and backthrust, while M2 corresponds to the Guttingkeng (Gtk) Formation, a mudstone unit overlying M1. The remaining host rocks are sandstones (ss). A 0.5-km-thick cap layer (C) covers the surface. Structural units: T1, T2, and T3 denote the forethrust, backthrust, and décollement, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

them as unyieldable.

To ensure the model represents realistic conditions, we selected frames for analysis based on the criteria of *no significant distortion of structural geometry* and *ambient deviatoric stress values within a reasonable range*. First, since the structural geometry of the model is based on the present-day profile, the model time is constrained so that total deformation does not significantly distort the geometry. Second, as model time progresses, the stress within the domain progressively increases. To define a reasonable upper limit for the simulation duration, we performed a first-order estimation of the maximum sustainable stress at a representative depth. Assuming lithostatic conditions, we approximate the normal stress (σ_n) at a depth (h) of 10 km using a rock density (ρ) of 2700 kg/m^3 and gravitational acceleration (g) of 9.8 m/s^2 :

$$\sigma_n \approx \rho gh \quad (6)$$

We then estimate the maximum shear stress ($\tau_{\max}$) using a simplified Mohr-Coulomb criterion, with cohesion (C) set to 40 MPa and a friction angle (ϕ) of 30° :

$$\tau_{\max} = C + \sigma_n \tan \phi \quad (7)$$

Based on these parameters, the estimated in-situ deviatoric stress limit at 10 km depth is approximately 200 MPa. We can then estimate the time (t) required to accumulate this stress level, assuming a shear modulus (G) of 30 GPa, a model width (L) of 100 km, and a boundary convergence rate (v) of 30 mm/yr:

$$t \approx \frac{\tau_{\max} \cdot L}{G \cdot v} = \frac{200 \times 10^6 \text{ Pa} \times 10^5 \text{ m}}{30 \times 10^9 \text{ Pa} \times 30 \times 10^{-3} \text{ m/yr}} \approx 2.22 \times 10^4 \text{ years} \quad (8)$$

This first-order estimate indicates that simulations extending beyond approximately 22,200 years would likely generate unrealistically high ambient stress and widespread plastic failure, in addition to compromising the initial structural geometry. Consequently, we confined our detailed analysis to the interval between 10,000 and 20,000 years. In the numerical model, results are saved at 400-year intervals as one frame. Our selected analysis window corresponds to frames 25 through 50. During this interval, deviatoric stress increases in response to compressive boundary loading while remaining within the physically reasonable bounds established above. The strain rate distribution is observed to progressively intensify (Fig. 4), initiating in the shallow portions of the thrusts and subsequently expanding to deeper detachments, eventually achieving a quasi-steady distribution.

We designed a series of experimental groups to investigate the effects of specific mechanical properties in combination with particular structural units. The first group, outlined in Fig. 5, focuses on lithological units of mudstone (M1) (Fig. 2). The mechanical behaviors examined include frictional behavior (M1f), constant viscous behavior (M1cv), Newtonian viscous behavior (M1nv), and non-Newtonian viscous behavior (M1nnv). The second group, as outlined in Fig. 6, comprises series models featuring mudstone units M2 and M12, as well as the complete system of all units (M12T123). This group incorporates mechanical properties of friction and non-Newtonian viscosity. The third

Viscosity and Strength Profiles for Different Viscous Models

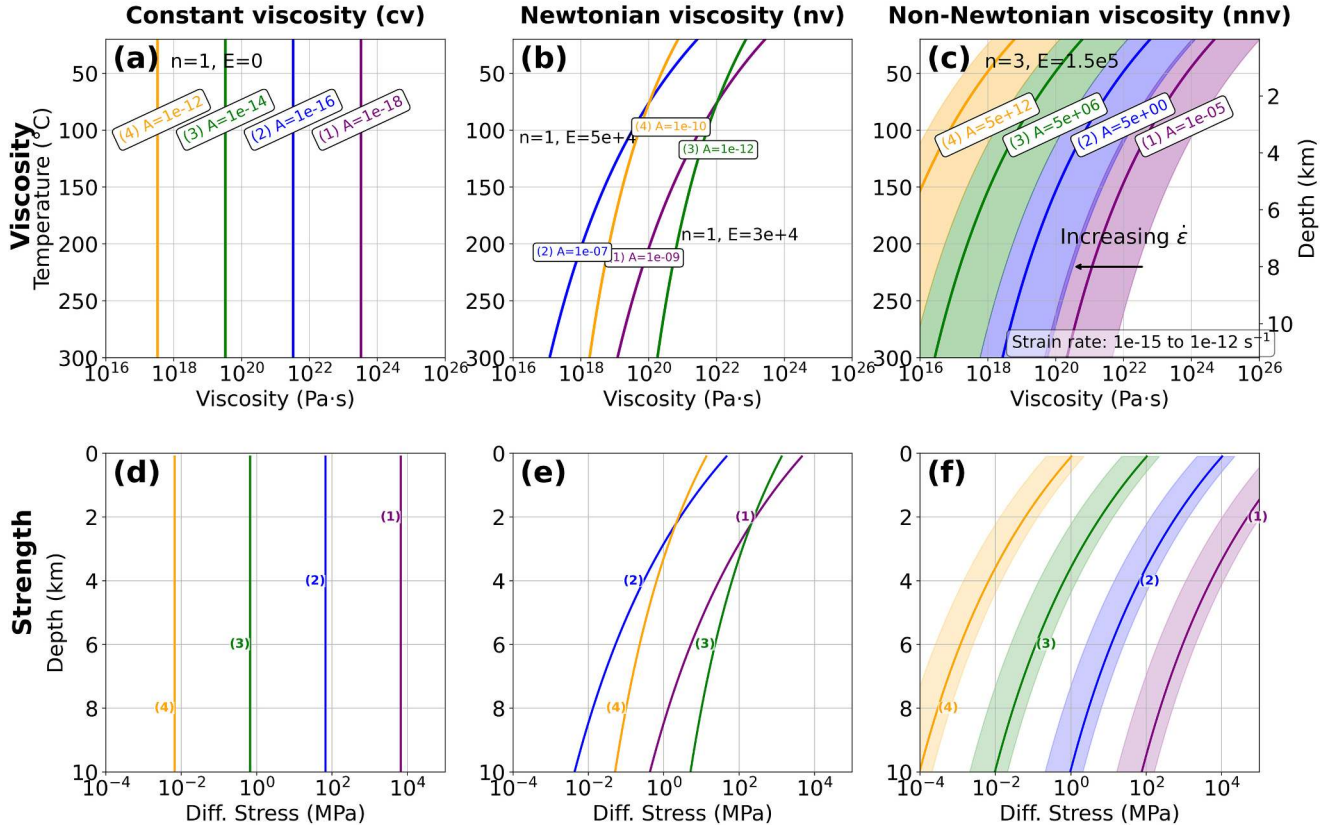


Fig. 3. Rheological profiles used in the numerical experiments. Top row (a–c): Shear viscosity η (log scale) plotted against temperature (left axis) and depth (right axis), governed by the flow law in Eq. (5). Bottom row (d–f): Corresponding strength profiles (differential stress) versus depth. The columns represent three viscous models: (a, d) Constant viscosity (cv) and (b, e) Newtonian viscosity (nv), which are both strain-rate independent. In contrast, (c, f) Non-Newtonian viscosity (nnv) is strain-rate dependent; the color-shaded regions in (c) and (f) illustrate the range of viscosity and strength variations corresponding to strain rates between 10 and 15 and 10–12 s^{−1}. The pre-exponential factor A is labeled on each curve, while the power-law index n and activation energy E are indicated in the panels.

group, detailed in Fig. 7, evaluates the effects of structural units. These units include forethrust (T1), backthrust (T2), detachment (T3), and their combinations, featuring either frictional or viscous behavior.

2.3. Model outputs and key observables

To analyze the model results for various parameter combinations, we present several key features. These include surface velocities (both vertical and horizontal components) from frames 25 to 50, (corresponding to 10,000 to 20,000 years) and strain rate specifically of frame 37 (at 14,800 years) as a representative. Key distinguishing features include the surface vertical-to-horizontal velocity ratio (V_v / V_h), the standard deviation (std), indicating variance, of the ratio, and surface deformation patterns. The V_v / V_h ratio is calculated as the peak-to-peak vertical velocities (V_v) divided by the peak-to-peak horizontal velocities (V_h) between $X = 37$ km and 67 km along the profile. In our model, the reference point ($x = 0$) is located at the left boundary; accordingly, negative V_h denotes leftward motion in the model, which corresponds to westward and foreland-directed motion in Southwestern Taiwan FTB, while positive V_h denotes the opposite (hinterland-directed) motion. Positive V_v denotes uplift, and negative V_v denotes subsidence. While some models show surface deformation focused only on narrower zones, we adopt a unified range to standardize the discussions across all models for consistency and simplicity. The std of V_v / V_h indicates the variability of V_v / V_h values across frames 25 to 50. Another key feature of the model is the surface deformation pattern, including the morphology of surface velocities, such as the symmetry or asymmetry of V_v . In cases of asymmetry, further analysis can focus on characteristics such as the

steepness of each limb of the velocity profiles. Deformation patterns also encompass horizontal velocity gradients ($\frac{\partial V_h}{\partial x}$), from which compres-

sional or extensional regimes linked to specific structures can be inferred. The relationship between V_v and V_h within a given regime further provides insight into the kinematic coupling of the system. These metrics, which describe deformation behaviors across different frames under varying ambient stress conditions, highlight the system's response to stress perturbations. Observations, such as those by Rau et al. (2022) suggest that surface velocities can fluctuate during events like earthquakes due to stress-induced triggering. Therefore, analyzing the std in velocity and related patterns is essential for understanding the dynamic responses of the investigated units with their specific mechanical properties.

3. Result

3.1. Experimental Group 1: Active M1

We begin our investigation of the first experimental group of models with active M1 unit, representing the high pore-pressure GtK Formation (Fig. 5 and Fig. 8). First, we consider M1 under plastic deformation. These models—M1f_1 through M1f_4 (Fig. 8a–d)—represent progressively higher pore pressure conditions, which effectively decreases their frictional angles to 32°, 24°, 16°, and 8°, respectively. The strain rate plots at frame 37 indicate that the deformation is concentrated along the top portion of M1, at the margins of a triangular area defined by the fore-

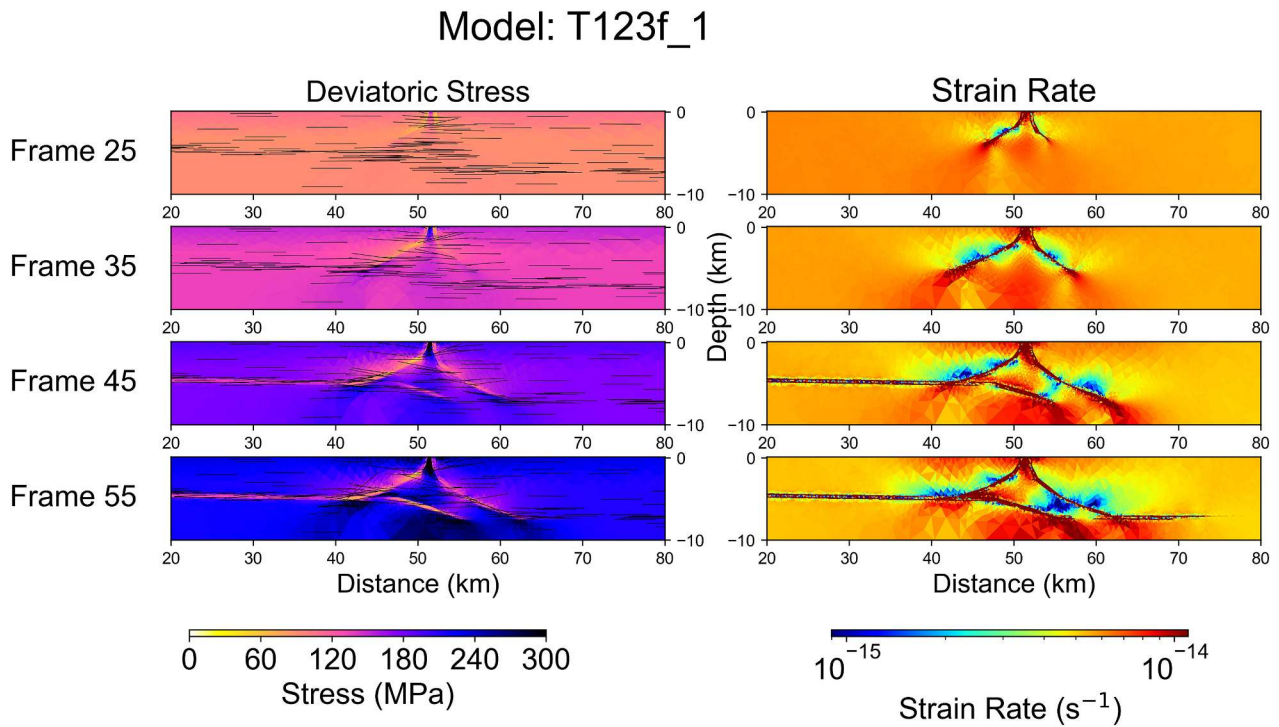


Fig. 4. Temporal evolution of stress and strain rate fields in model T123f_1 from frames 25 to 55. Each frame shows two panels: deviatoric stress (MPa) with maximum (most compressive) principal stress (σ_1) orientations indicated by black bars (left panel), and strain rate (s^{-1}) distribution (right panel).

Experimental Group 1

Unit	Models (f)	Mean	Std	Models (cv)	Mean	Std	Models (nv)	Mean	Std	Models (nnv)	Mean	Std
M1	M1f_1 (a)	0.354	0.154	M1cv_1 (e)	0.004	0.002	M1nv_1 (i)	0.222	0.020	M1nnv_1	0.010	0.002
	M1f_2 (b)	0.519	0.045	M1cv_2 (f)	0.196	0.029	M1nv_2	0.554	0.013	M1nnv_2 (j)	0.286	0.026
	M1f_3 (c)	0.566	0.004	M1cv_3 (g)	0.594	0.004	M1nv_3	0.159	0.022	M1nnv_3	0.597	0.003
	M1f_4 (d)	0.583	0.004	M1cv_4 (h)	0.596	0.005	M1nv_4	0.573	0.010	M1nnv_4 (k)	0.596	0.005

0.0 0.5 1.0 0.0 0.2
Mean Std

Fig. 5. Experimental Group 1: Model configuration and surface deformation statistics for the M1 series models. Columns define the mechanical properties (setup parameters, details see Fig. 3), while rows list model names with corresponding figure references (Fig. 8). The color-coded bars below each model name display the experimental results: blue hues indicate the mean V_v/V_h ratios, and red hues indicate the standard deviation (std). The color saturation corresponds to the magnitude of the value, where darker shades indicate higher values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and back-thrusts. Strain rates in low friction models are higher and extend to a greater depth. The vertical and horizontal velocity profiles at the surface (V_v and V_h) exhibit similar patterns across all models: a strong uplift and a broad horizontal shortening (red-shaded background in the V_h plot) with a 3-km wide extension zone (blue-shaded background). The extensional zone and maximum V_v occurs above the central tip of M1, suggesting that unit geometry plays a critical role in shaping surface deformation patterns. Additionally, the experimental results reveal that decreasing friction angles lead to higher mean V_v/V_h ratios (increasing from ~ 0.4 in M1f_1 to ~ 0.58 in M1f_4) and lower variance (smaller std) in V_v/V_h ratios, indicating that an active M1 would produce greater uplift regardless of stress state. On the other hand, high-friction models exhibit greater variance in V_v/V_h ratios, with vertical velocities showing more significant sensitivity to stress variations. This occurs because elements in high-friction models yield only under high stress, which occur only sporadically.

We next consider an active M1 unit under viscous deformation. Our observations show that viscosity magnitude is the primary control on

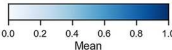
deformation patterns, regardless of the specific viscosity formulation used. When viscosity exceeds 10^{21} Pa·s, deformation is minimal. This pattern holds across all viscosity types—constant viscosity (cv) (Fig. 8e–h), Newtonian viscosity (nv), and non-Newtonian viscosity (nnv) models. For example, in model M1cv_2 (Fig. 8f), both strain rates and mean V_v/V_h ratios remain low (~ 0.2). Similarly, models M1nv_1 and M1nnv_2 (Fig. 8i, j), where bulk viscosity exceeds this threshold, show limited deformation. In contrast, low viscosity models (e.g., M1nnv_4) (Fig. 5 and Fig. 8k) exhibit strain rates concentrated at the top margins of the triangular M1 unit, though deformation is more evenly distributed within the unit compared to frictional models (Figs. 8a–d).

The absolute viscosity range controls deformation intensity, while depth-dependent or strain rate-dependent viscous behaviors have minimal impact on surface velocity and deformation patterns. Given this consistency across viscosity formulations, we use non-Newtonian viscosity (nnv) (Fig. 3c) as representative viscous behavior for subsequent models, rather than presenting details for each individual viscous case.

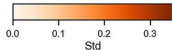
Viscous materials, particularly stronger ones, exhibit lower variance

Experimental Group 2

Unit(s)	Models (f)	Mean	Std	Models (nnv)	Mean	Std
M2	M2f_1	0.339	0.069	M2nnv_1	0.002	0.002
	M2f_2 (a)	0.449	0.055	M2nnv_2	0.011	0.004
	M2f_3	0.504	0.037	M2nnv_3 (c)	0.148	0.014
	M2f_4 (b)	0.543	0.071	M2nnv_4 (d)	0.483	0.039
M12	M12f_1	0.369	0.091	M12nnv_1	0.003	0.003
	M12f_2 (e)	0.461	0.050	M12nnv_2	0.011	0.004
	M12f_3	0.514	0.034	M12nnv_3 (g)	0.145	0.012
	M12f_4 (f)	0.573	0.042	M12nnv_4 (h)	0.678	0.079
M12T123	M12T123f_1	0.429	0.146	M12T123nnv_1	0.002	0.003
	M12T123f_2 (i)	0.571	0.161	M12T123nnv_2	0.014	0.004
	M12T123f_3	0.761	0.314	M12T123nnv_3 (k)	0.166	0.014
	M12T123f_4 (j)	0.632	0.110	M12T123nnv_4 (l)	1.001	0.194



Mean



Std

Fig. 6. Experimental Group 2: Model configuration and surface deformation statistics for M2, M12, and M12T123 series models. Columns define the mechanical properties (setup parameters), while rows correspond to the targeted structural units (M2, M12, and M12T123) with associated profile references (Fig. 9). The color-coded bars below each model name display the experimental results: blue hues indicate the mean V_v/V_h ratios, and red hues indicate the standard deviation (std). The color saturation corresponds to the magnitude of the value, where darker shades indicate higher values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in surface velocities compared to frictional (M1f) series models (Fig. 5), suggesting reduced sensitivity to stress perturbations. As viscosity decreases, both V_v and V_v/V_h ratios increase, indicating higher material mobility. The surface deformation patterns between weaker frictional and viscous materials are remarkably similar. Specifically, M1f_4 and M1nnv_4 (Fig. 8d, k) both display predominantly symmetrical deformation patterns with mean V_v/V_h ratios of approximately 0.6, despite their fundamentally different deformation mechanisms.

3.2. Experimental Group 2: Active M1 and M2

We next investigate the second experimental group of models incorporating active M1 and M2 units, representing the Gtk formation (Fig. 6 and Fig. 9). Our analysis begins with active M2 units, focusing on mudstone strata adjacent to M1, then examines results for combined active M1 and M2 units (M12), and subsequently evaluates models integrating M12 with all thrust structures (M12T123).

The M2f_1 to M2f_4 models (Fig. 6 and 9a–b) investigate M2 unit with frictional rheology, while M2nnv_1 to M2nnv_4 (Fig. 6 and 9c–d) evaluate M2 unit with viscous rheology. Strain rate localization in frictional models occurs along the upper margins of M2 units, whereas viscous models exhibit uniformly distributed strain rates with preferential concentration in the right M2 unit relative to the left. A characteristic feature of M2-induced surface deformation comprises two asymmetrical humps in V_v . The right-side peak exhibits greater amplitude than the left-side peak, with V_v maxima spatially coincident with V_h extensional zones. In the lowest friction case (M2f_4), the mean V_v/V_h ratio reaches ~ 0.5 .

The M12 series models (Fig. 6 and 9e–h) with inactive thrusts demonstrate that incorporation of the additional mobile M1 unit moderately increases V_v/V_h ratios beyond the contribution of M2 alone (e.g., rising to ~ 0.57 in M12f_4). Frictional models (M12f_1 to M12f_4; Fig. 9e–f) produce deformation patterns analogous to the M2f series, characterized by two distinct velocity humps accompanied by V_h extension. A relatively subdued depression between the two peaks results from M1's contribution of minor vertical uplift. Viscous models (M12nnv_1 to M12nnv_4; Fig. 9g–h) generate nearly symmetrical V_v distributions with minor V_h extension. These models indicate that when M1 and M2 units contain similar rheological properties, surface

deformation is predominantly controlled by M2. Given that observed surface V_v exhibits asymmetrical patterns without dual humps or enhanced right-side amplitudes, we infer that M2 cannot be the dominant active unit in reality.

Next, we examine models incorporating both thrusts and detachment (M12T123; Fig. 9i–l). Frictional models (M12T123f_1 to M12T123f_4; Fig. 9i–j) localize strain rates within the upper portions of active units, consistent with the pattern observed in the M2f and M12f series. These configurations produce substantial variance in both V_v and V_h , with V_v displaying a single, approximately symmetrical hump and V_h exhibiting smoothly varying compressional characteristics. The mean V_v/V_h ratio in the lowest friction model (M12T123f_4) reaches ~ 0.6 . Viscous models (M12T123nnv_1 to M12T123nnv_4; Fig. 9k–l) concentrate strain rates within the lower portions of active units. Although V_v maintains a single hump comparable to equivalent frictional models (with a remarkably high V_v/V_h of ~ 1.0 in M12T123nnv_4), V_h demonstrates lower velocity gradients and reduced velocity variance. This observation suggests that V_h patterns, in some cases, may serve as diagnostic indicators for inferring the mechanical properties of active geological units.

3.3. Experimental Group 3: Active T1, T2, and T3

In the following section, we examine the third experimental group (Fig. 7 and Fig. 10) focusing on the role of structural units—forethrust, backthrust, and detachment—and their frictional behavior. For models with an active frictional forethrust (T1f_1 to T1f_4; Fig. 7 and 10a–b), strain rates increase and become concentrated along the central segment of the forethrust as friction decreases. Both V_v and the V_v/V_h ratios gradually increase while variance decreases as frictional strength weakens, with the V_v/V_h ratios reaching approximately 0.5 due to the single forethrust geometry. The resulting deformation pattern exhibits an asymmetrical V_v distribution, characterized by broad, gently-varying right flank and a narrower, steeper left flank. When models incorporate an active detachment in addition to the forethrust (T13f_1 to T13f_4; Fig. 7 and 10c–d), strain rate still concentrates along the forethrust. Strain rate along the detachment begins to increase only when friction is sufficiently reduced. This leads to a further increase in the mean V_v/V_h ratio, reaching up to ~ 0.7 in the T13f_4 model. However, the overall

Experimental Group 3

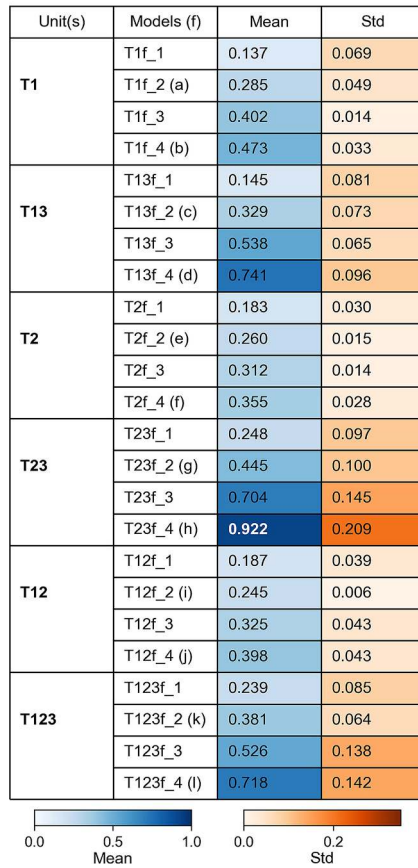


Fig. 7. Experimental Group 3: Model configuration and surface deformation statistics for frictional T series models. Columns define the mechanical properties (setup parameters), while rows correspond to the targeted structural units (T1 to T123) with associated profile references (Fig. 10). The color-coded bars below each model name display the experimental results: blue hues indicate the mean V_v/V_h ratios, and red hues indicate the standard deviation (std). The color saturation corresponds to the magnitude of the value, where darker shades indicate higher values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

deformation pattern remains similar to models without an active detachment, albeit with higher variance. This increased variance reflects the fact that detachments respond to varying stress scenarios in a more unstable and dynamic manner. Those models with forethrust but without backthrust experience significant transitions from extension to compression over a short distance (~ 3 km), a scale that might be challenging to observe geodetically.

Models T2f_1 to T2f_4 (Fig. 7 and 10e–f) feature only an active backthrust. Strain rates concentrate at the central segment of the thrust, similar to forethrust models, but exhibit smaller V_v and V_v/V_h ratios compared to models T1f_1 to T1f_4. The mean V_v/V_h ratio in T2f_4 reaches only ~ 0.4 , lower than the ~ 0.5 in T1f_4. The T2f series also shows slightly smaller V_v/V_h variance and an asymmetrical V_v , characterized by a steep right flank and a broader left flank. When the backthrust combines with an active detachment (T23f_1 to T23f_4; Fig. 7 and 10g–h), reducing frictional strength produces a more pronounced V_v/V_h increase than T13f models. In T23f_4, the mean V_v/V_h reaches ~ 0.9 , compared to only ~ 0.7 in the equivalent model T13f_4. The T23f series also exhibits greater V_v and V_v/V_h variance than the T13f series. This suggests that, while the forethrust and backthrust geometry may appear as mirror images of each other, the V_v magnitudes and variances indicate that fault geometry, dipping direction, and the configuration of

structural units significantly influence surface deformation characteristics. Furthermore, the magnitudes of V_v or V_v/V_h are not simply the linear sum of individual fault contributions.

Next, we examine Model T12 (Fig. 7), which represents the configuration with both active forethrust and backthrust. In the T12f series (T12f_1 to T12f_4; Fig. 7 and 10i–j), higher frictional strength initially leads to greater strain rates in the backthrust (Fig. 10i), resulting in a surface deformation peak slightly higher than that of the forethrust. However, as frictional strength decreases, forethrust strain rates exceed those of the backthrust, shifting the deformation peak such that the forethrust side exhibits slightly higher V_v than the backthrust side (Fig. 10j). Overall, mean V_v/V_h ratios in these models (reaching ~ 0.4 in T12f_4) remain lower than in the T13f or T23f series.

The T123f series (Fig. 7 and 10k–l) extends the T12f configuration by incorporating a detachment alongside both thrusts. Compared to T12f models, detachment inclusion increases V_v , V_v/V_h , and their variance. With decreasing frictional strength from T123f_1 to T123f_4, the detachment becomes increasingly active, producing a pronounced V_v/V_h increase from ~ 0.2 to ~ 0.7 . A notable pattern emerges in low-friction models (e.g., T123f_4; Fig. 10l): deformation initially exhibits two distinct peaks under low ambient stress that gradually merge into a single peak as stress increases. This demonstrates the strong correlation between deformation characteristics and stress conditions. Additionally, all T123f models display minor V_h extensions corresponding to V_v peak values.

4. Discussion

4.1. Comparison with field observations and implications for active deformation

This study provides a preliminary framework for characterizing which structural units are most likely active. While we do not attempt to directly fit geodetic observations, our results offer a valuable reference for comparison with field data. To facilitate this comparison, we have plotted the geodetic observations along profile G–G' from Le Béon et al. (2017) as red lines in the vertical velocity panels of Fig. 11. These observations reveal an asymmetrical V_v velocity profile and a V_v/V_h ratio of ~ 0.7 . The V_v exhibits a narrow, steep gradient localized between the two thrusts, transitioning to a broad, gentle gradient toward the foreland. This spatial pattern reflects the asymmetric V_v of the active units.

Additionally, observed coseismic surface velocity fluctuations (Rau et al., 2022) indicate sensitivity to ambient stress variations. Since frictional models display higher variance under changing stress conditions than viscous models (Figs. 5 and 6), we infer that the active unit likely possesses frictional properties. Spatially, this deformation pattern closely resembles that of our T23f_4 model, dominated by an active backthrust and detachment (Fig. 10h). Our model interpretation suggests that the gentle gradient toward the foreland results from active backthrust interseismic creep, while the minor hinterland gradient reflects an active detachment ramp. The observed V_v/V_h ratio corresponds closely to our T23f_3 (Fig. 11a) to T23f_4 (Fig. 11b) model range (Fig. 7), indicating friction angles between approximately 8° and 16° if this structural configuration is accurate. However, the potential contribution of other units (e.g., M1) to the overall deformation cannot be excluded. For example, Fig. 11 (c–f) displays the combined effects of active M1 with viscous behavior, and frictional backthrust and detachment. Models T23f_3M1nnv_2 and T23f_3M1nnv_3 (Fig. 11c, d) demonstrate the contribution of viscous M1 in addition to T23f_3 (Fig. 7, $V_v/V_h = 0.704 \pm 0.145$), while models T23f_4M1nnv_2 and T23f_4M1nnv_3 (Fig. 11e, f) demonstrate the contribution of viscous M1 in addition to T23f_4 (Fig. 7, $V_v/V_h = 0.922 \pm 0.209$). As the plotted observations generally align with the model curves (Fig. 11) and show no significant differences from backthrust-and-detachment-only models, we conclude that backthrust and detachment are the primary contributors to surface deformation.

Experimental Group 1: Testing M1 with varying rheology

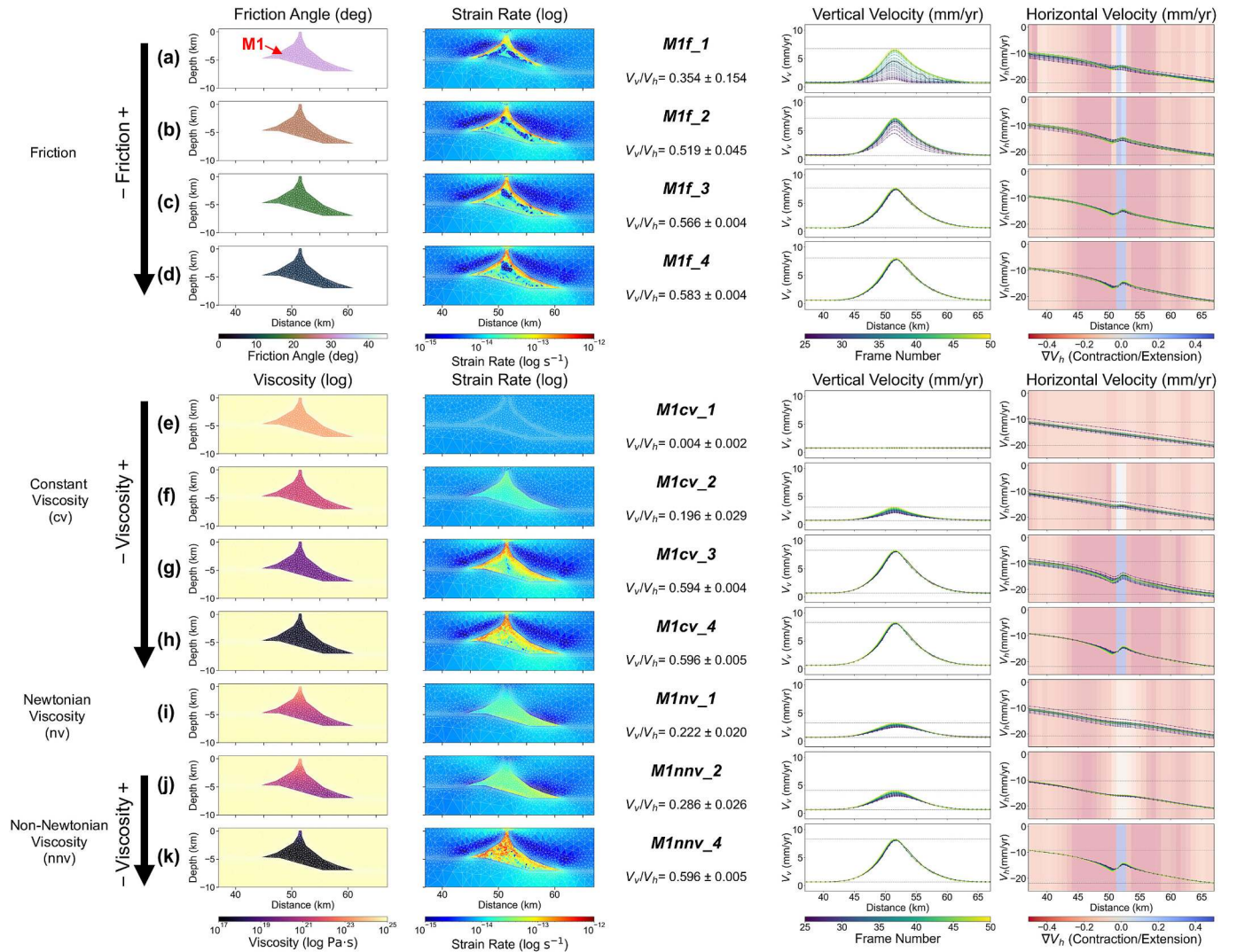


Fig. 8. Experimental Group 1 – Active M1 unit models testing different rheologies. Panels (a–d) show frictional models; (e–h) show constant viscosity models; (i) shows the Newtonian viscosity model, selected as a representative case; and (j–k) show non-Newtonian models (high-viscosity M1nnv_2 and low-viscosity M1nnv_4, respectively). Each row displays: (1) rheological property profiles (friction angle or viscosity); (2) logarithmic strain rate; (3) vertical surface velocity profiles; and (4) horizontal surface velocity profiles plotted over the horizontal gradient ∇V_h (background color); positive values indicate rightward motion and negative values indicate leftward motion. Statistics for the V_v / V_h ratio are listed in the center column. The cross-sections (left two columns) are snapshots at frame 37, while the velocity profiles (right two columns) show the evolution from frame 25 to 50.

4.2. Lithological or structural controls on surface deformation?

Previous studies have attributed unmodeled anomalously high surface deformation to lithological controls, invoking the term “mud diapir (ism)” to suggest that viscous behavior drives aseismic deformation with an unconstrained pattern (e.g., [Ching et al., 2016](#)). While lithology profoundly influences surface deformation characteristics, the mechanisms through which it operates are varied and function on multiple scales.

Direct lithological influence occurs through mechanical properties that affect slip behavior. Phyllosilicate mineral content directly controls frictional strength and permeability ([Moore and Lockner, 2011](#); [Takahashi et al., 2014](#); [Tembe et al., 2010](#); [Zhang et al., 2019](#)). Higher phyllosilicate concentrations produce low friction values (~ 0.1 to 0.3). These low friction conditions correlate with creeping fault segments, as observed on the Chihshang fault in eastern Taiwan (e.g., [Collettini et al., 2019](#); [Den Hartog et al., 2021](#); [Thomas et al., 2014](#)) and the San Andreas Fault ([Carpenter et al., 2012](#); [Moore and Rymer, 2007](#)).

Indirect lithological influence operates through structural style development, where the asymmetrical high-uplift anomaly pattern observed here appears primarily related to underlying backthrust activity. Structural architecture is influenced by lithological rheology over geological timescales, as demonstrated in regions with salt, mudstone, or shale substrates through field observations and modeling studies (e.g., [Célini et al., 2024](#); [Cotton and Koyi, 2000](#); [Dean et al., 2015](#); [Huang et al., 2020](#); [Luján et al., 2003](#); [Ruh et al., 2012](#); [Simpson, 2009](#); [Tămaş et al., 2020](#); [Von Hagke et al., 2024](#)). This lithological control on structure is evident in Taiwan's contrasting fold-and-thrust belts: northern Taiwan primarily comprises forethrusts ([Yang et al., 2016](#); [Yue et al., 2011](#); [Yue et al., 2005](#)), whereas southwestern Taiwan features prominent backthrusts inferred from seismic reflection and surface geological observations ([Aleem et al., 2025](#); [Huang et al., 2004](#); [Le Béon et al., 2024](#); [Le Béon et al., 2019](#); [Le Béon et al., 2017](#)). The study profile exhibits backthrusts and forethrusts connected at their apex, forming oppositely dipping strata ([Fig. 1b, c](#)), a configuration that preferentially develops above mudstone rather than sandstone substrates, as

Experimental Group 2: Testing M2 with varying rheology and active units

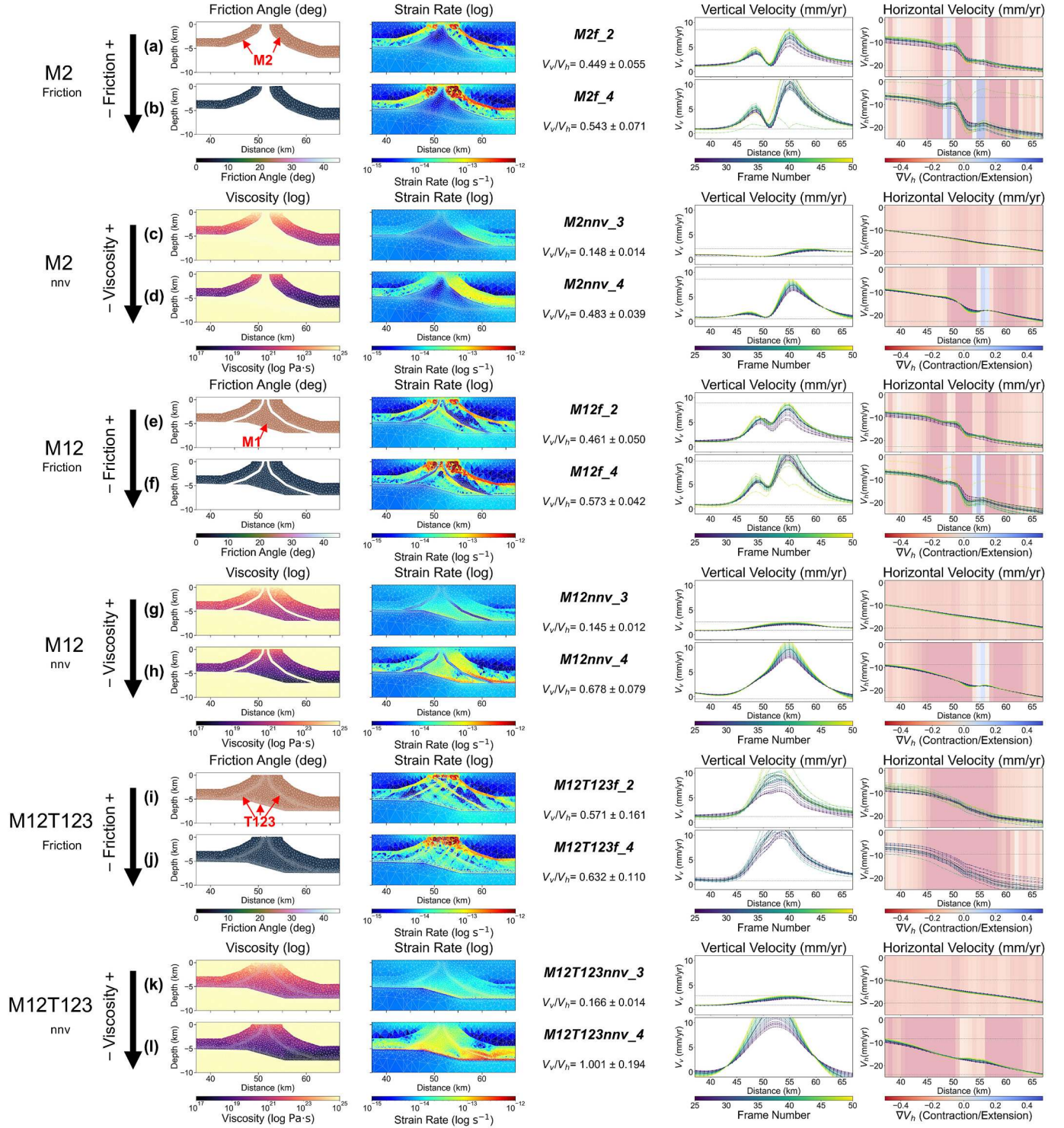


Fig. 9. Experimental Group 2 – Testing active units (M2, M12, M12T123) with varying rheologies. Panels (a–b) display M2 frictional models; (c–d) M2 non-Newtonian models; (e–f) M12 frictional models; (g–h) M12 non-Newtonian models; (i–j) M12T123 frictional models; and (k–l) M12T123 non-Newtonian models. In each pair, the models compare different rheological strength parameters. Each row displays: (1) rheological property profiles (friction angle or viscosity); (2) logarithmic strain rate; (3) vertical surface velocity profiles; and (4) horizontal surface velocity profiles plotted over the horizontal gradient ∇V_h (background color); positive values indicate rightward motion and negative values indicate leftward motion. Statistics for the V_v/V_h ratio are listed in the center column. The cross-sections (left two columns) are snapshots at frame 37, while the velocity profiles (right two columns) show the evolution from frame 25 to 50.

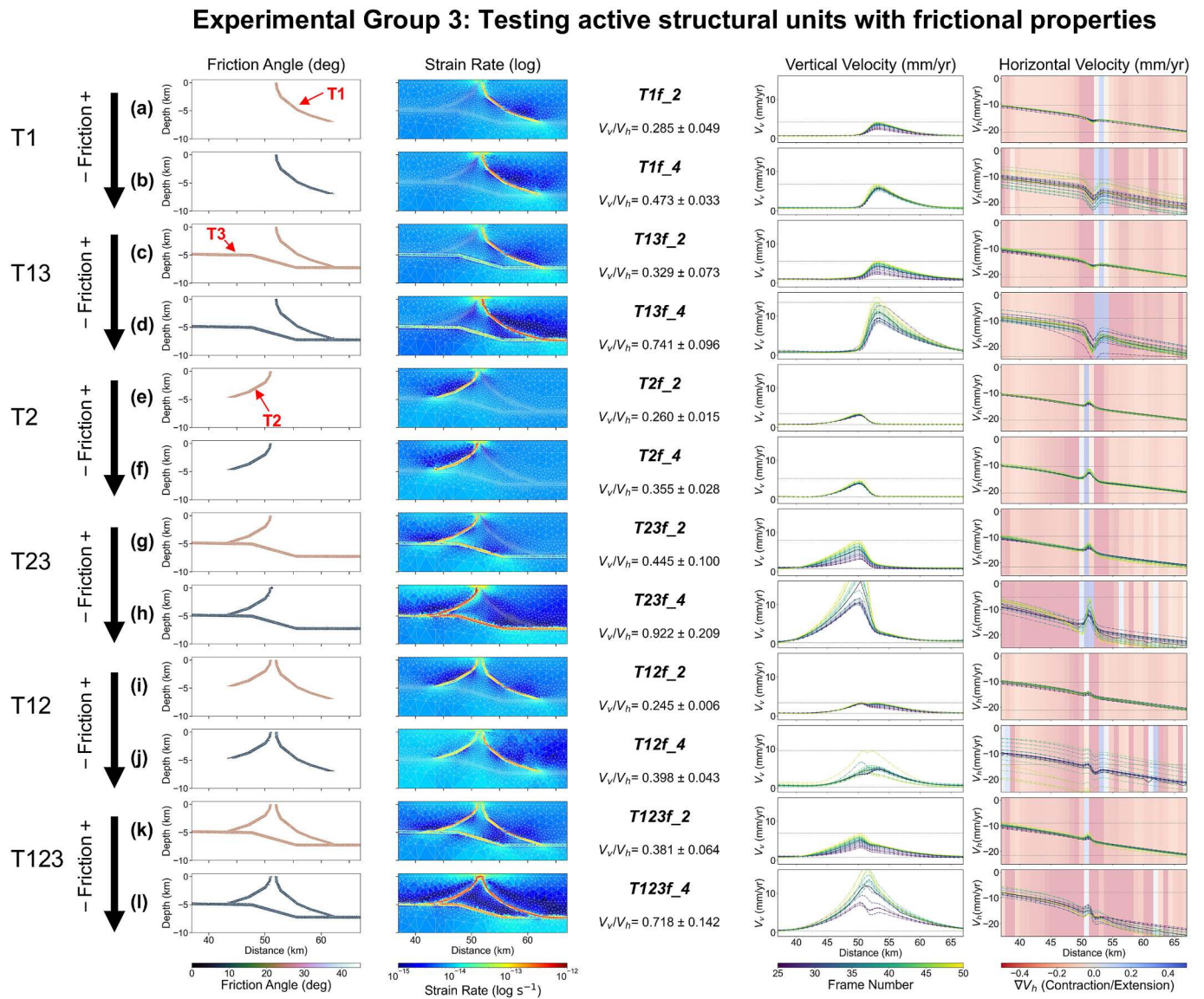


Fig. 10. Experimental Group 3 – Testing active structural units (T1, T13, T2, T23, T12, T123) with frictional properties. Panels (a–b) display configuration T1; (c–d) configuration T13; (e–f) configuration T2; (g–h) configuration T23; (i–j) configuration T12; and (k–l) configuration T123. In each pair, the models compare different frictional parameters. Each row displays: (1) rheological property profiles (friction angle); (2) logarithmic strain rate; (3) vertical surface velocity profiles; and (4) horizontal surface velocity profiles plotted over the horizontal gradient ∇V_h (background color); positive values indicate rightward motion and negative values indicate leftward motion. Statistics for the V_v / V_h ratio are listed in the center column. The cross-sections (left two columns) are snapshots at frame 37, while the velocity profiles (right two columns) show the evolution from frame 25 to 50.

demonstrated by discrete element simulations (Dean et al., 2015). Based on this study, structural geometry primarily controls first-order deformation patterns, while surface velocity magnitudes reflect strength variations closely tied to lithology. Thus, lithology in the FTB of SW Taiwan could exert diverse direct and indirect influences on surface deformation across multiple timescales, requiring careful consideration of each mechanism and its specific effects.

4.3. Limitation of this study

Previous studies and models (e.g., Ching et al., 2016; Kuo, 2017) have faced challenges in exploring diverse deformation mechanisms, constructing realistic fault geometry, or accounting for surface deformation caused by rock masses outside the fault zones. This study addresses these gaps by utilizing finite-element method to analyze surface deformation caused by shallow crustal regions, offering insights that could be extended to other geological profiles and regions. Despite these

advancements, our study is not without limitations. Our model findings highlight the critical role of structural geometry in shaping deformation patterns. For example, forethrust, backthrust, or detachment activity produces varying V_v / V_h ratios, influenced by the thrusts' dipping direction, angle, and the presence and the position of a ramp; additionally, during deformation of the mudstone layer (M1), strain localization occurs at its edges, regardless of whether the behavior is frictional or viscous. This demonstrates that geometric configurations play a crucial role in determining the distribution of localized strain, which, in turn, impacts surface deformation patterns. However, this study focuses on predicting model results for fault geometry based on a specific tectonic profile (Le Béon et al., 2017). Given the known lateral variations in structural geometry within this mudstone-rich area, the generalizability of our findings to other along-strike profiles in the region may be limited. To address this, future studies should incorporate structural geologists to explore a broader range of geometries by varying fault dip angles, ramp positions, and other structural parameters within

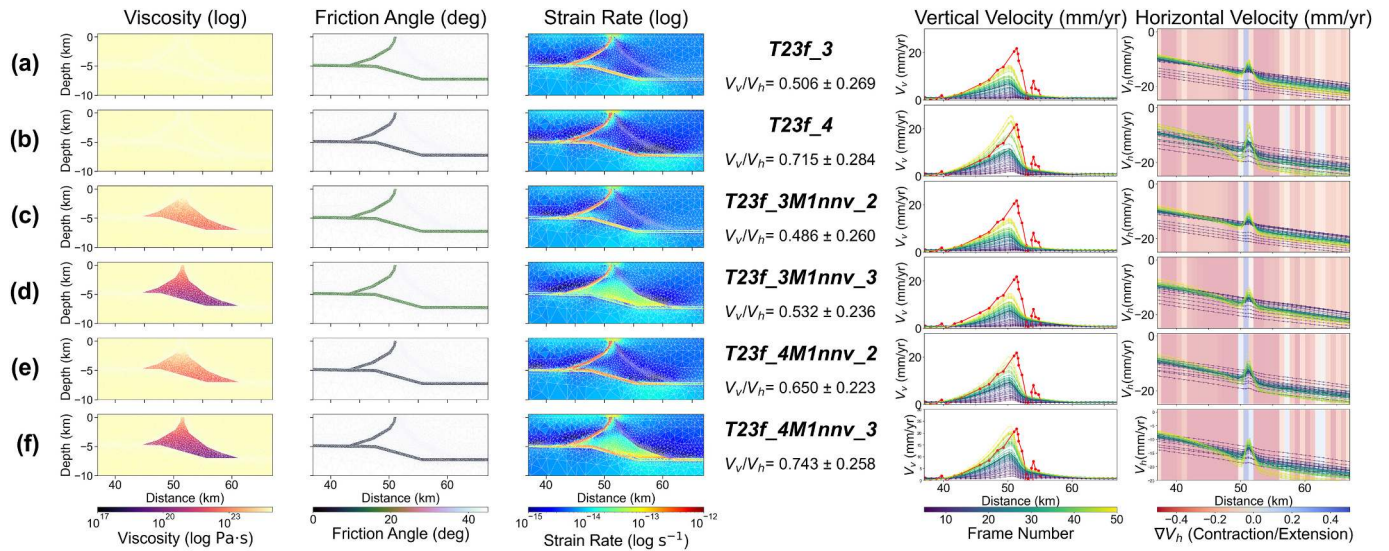


Fig. 11. Comparison of models combining the active backthrust/detachment (T23) and M1 unit deformation. Columns display (from left to right) viscosity, friction angle, strain rate, vertical velocity (V_v), and horizontal velocity gradient (∇V_h). (a, b) Reference models showing the T23 structure only. (c, d) Models combining T23f_3 (16° friction angle) with a viscous M1 unit (nnv2 and nnv3). (e, f) The same M1 configurations combined with T23f_4 (8° friction angle). The vertical velocity plots include observed leveling data (red line), as shown in Fig. 1c, for comparison. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

geologically reasonable bounds. Such efforts will allow for a more comprehensive understanding of how structural uncertainty influences model predictions, and will enable iterative refinement of fault models based on observational constraints. Furthermore, the models in this study only account for specific combinations of units and mechanical properties. The mechanical properties are simplified, with uniform parameters applied to all targeted units within a model. The study considers only two end-member behaviors—frictional (elastoplastic) and viscous—appearing individually in each model, with parameter values limited to four specific test cases. However, real-world conditions are likely far more complex. In reality, the mechanical properties of each active unit may differ, and behaviors may involve a mixture of elastoplastic and viscous characteristics (frictional-viscous flow in fault zones) (e.g., Bos and Spiers, 2002).

Additionally, different units typically exhibit varying factors influencing mechanical properties, such as mineral compositions, microstructures (e.g., Collettini et al., 2009; Yan et al., 2021), and permeability (e.g., Michie et al., 2021; Rawling et al., 2001). Even within identical lithological units, spatial heterogeneity can occur due to factors such as fluid presence, which can significantly weaken materials (e.g., Griggs, 1967; Chen et al., 2025a; Chen et al., 2025b). This study does not consider fluid flux, pore-fluid pressure, or fluid diffusion effects, though field observations suggest substantial spatiotemporal fluid variations. For example, groundwater level anomalies have been documented (Wang et al., 2024) and post-seismic V_p / V_s changes, attributed to transient fluid migration, have been observed (Hsu et al., 2020). Similarly, geodetic observations have reported prolonged post-seismic enhanced strain rates (Lu et al., 2025), potentially linked to fluid-induced changes in mechanical properties. These mechanisms were not accounted for in the current models. Future, more realistic simulations should incorporate the effects of fluid diffusion and related processes to better represent real-world deformation behavior.

4.4. Implications to seismic hazards

The dominance of aseismic strain release does not imply that seismic hazards are eliminated. Instead, elastic strain may accumulate or be released at deeper levels and in neighboring areas, potentially affecting surrounding regions. For example, the 2016 Meinong earthquake

occurred on a deeper fault with no surface expression, yet caused strong ground motion due to directivity and site effects (e.g., Jian et al., 2017; Kanamori et al., 2017; Lee et al., 2016; Lin et al., 2018; Wen et al., 2020) and triggered surface deformation in mudstone-rich areas (e.g., Le Béon et al., 2017; Lin et al., 2024; Rau et al., 2022; Tsukahara and Takada, 2018), representing unexpected seismic-related hazards. This study further highlights that mudstone exhibiting viscous behavior with low viscosity could result in significant surface deformation, while friction-dominated materials can respond severely to stress perturbations. Therefore, understanding the mechanical properties of regional geological materials and structural geometry is essential, along with intensive monitoring of spatiotemporal surface deformation and regional physical conditions—such as subsurface fluid variations and in-situ stress. Moreover, integrating such data with numerical models, as demonstrated here, can provide a comprehensive understanding of active structures and the dominant mechanisms driving surface deformation, thereby enhancing hazard assessment.

5. Conclusions

This study explored the origin of anomalous aseismic uplift in the southwestern Taiwan fold-and-thrust belt through two-dimensional elasto-visco-plastic experiments. By varying the rheology of mudstone units and activating different fault geometries, we tested the competing roles of lithological weakening and structural architecture. Our results show that structural controls dominate uplift geometry. In particular, models incorporating active backthrusts and detachment ramps successfully reproduce the observed asymmetric V_v and surface uplift patterns, characterized by a broad western flank and steep eastern flank. While lithological weakening enhances deformation, it cannot alone explain the observed V_v asymmetry and the anomalously high V_v / V_h ratio. This indicates that surface signals often attributed to mud diapirism may instead arise from the combined activity of backthrusts and detachments. These findings provide diagnostic patterns to guide interpretation of geodetic data and highlight the importance of considering structural geometry when evaluating aseismic strain in mudstone-rich belts. Although simplified and restricted to a single profile, the models establish a framework for integrating geometry, rheology, and geodetic observations, offering a foundation for future studies that

incorporate multiple profiles.

Code availability

The numerical simulations were performed using the open-source software DynEarthSol (commit 5420830, dated November 1, 2023), which is available at <https://github.com/GeoFLAC/DynEarthSol> and archived on Zenodo (DOI: [10.5281/zenodo.21124059](https://doi.org/10.5281/zenodo.21124059)).

CRediT authorship contribution statement

Wan-Lin Hu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eh Tan:** Writing – review & editing, Validation, Supervision, Software, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was produced in this research.

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